



What to Do With Deauville 3 Interim Positron Emission Tomography/Computed Tomography Response in a Young Woman With Classic Unfavorable Hodgkin's Lymphoma: To Irradiate or Not to Irradiate?

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A 29-year-old woman with stage IIBX classic Hodgkin's lymphoma presented with a 5.1×7.6 -cm (mass to mediastinal ratio 0.34) anterior mediastinal mass and additional involvement of the subcarinal, posterior mediastinal, and right infraclavicular lymph nodes (Fig. 1). The patient was otherwise healthy, a lifelong nonsmoker, and European Cooperative Oncology Group

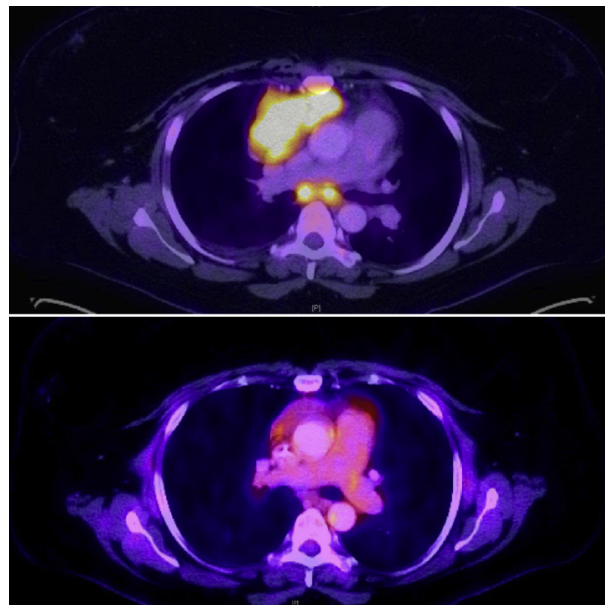


Fig. 1. Positron emission tomography/computed tomography imaging of Hodgkin's lymphoma, visualizing the anterior mediastinal mass and subcarinal lymph nodes. Before chemotherapy (top) and after 2 cycles of adriamycin, bleomycin, vinblastine, and dacarbazine (bottom), demonstrating Deauville 3 response with maximal standard uptake value of 2.9 for the anterior mediastinal mass, mediastinal blood pool 2.6, and liver 3.2.

performance status 0. She had a National Comprehensive Cancer Network International Prognostic Score of 1. She received 2 cycles of adriamycin, bleomycin, vinblastine, and dacarbazine (ABVD), and interim positron emission tomography demonstrated Deauville 3 response (Fig. 1).

Question

1. What is your treatment recommendation?
 - a. Intensify treatment to 2 cycles of escalated bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and prednisone for a total of 4 cycles, followed by involved-site radiation therapy (ISRT) to 30-36 Gy.
 - b. Continue with a total of 4 cycles of ABVD, followed by ISRT to 30-36 Gy.
 - c. Continue with a total of 6 cycles of ABVD without ISRT.
 - d. Continue with a total of 6 cycles of ABVD, followed by ISRT to 30-36 Gy.
 - e. Continue with a total of 4 cycles of AVD without ISRT.

See expert opinions on page 15.

What would you do? Follow the discussion on Twitter at #gyzone, and take the poll at www.redjournal.org/poll.

Note—CME is available for this feature as an ASTRO member benefit, to access visit <https://academy.astro.org>.

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GRAY ZONE EXPERT OPINIONS

Keep the Dose Low. Keep the Fields Tight



This is a common scenario.¹ The goal: maximize freedom from progression but minimize late risks. For a young woman, the main potential late risks secondary to radiation are breast cancer and cardiovascular disease. The main potential late risks from the escalation of chemotherapy are secondary myelodysplastic syndrome and infertility.

Her interim positron emission tomography (PET) Deauville score was 3, considered positive in the EORTC (European Organization for Research and Treatment of Cancer)/LYSA (Lymphoma Study Association)/FIL (Fondazione Italiana Linfomi) H10-U trial² but negative in the UK RATHL (response-adapted therapy for advanced Hodgkin lymphoma) trial.³

- a) reflects H10-U. Per this protocol, chemotherapy would be intensified to 2× BEACOPP (bleomycin, etoposide, Adriamycin, cyclophosphamide, Oncovin, procarbazine, prednisone)-escalated plus radiation therapy (RT), which was superior to ABVD (Adriamycin, bleomycin, vinblastine, dacarbazine; total 6) plus RT. However, this would expose her to the risks of myelodysplastic syndrome and infertility, and the National Comprehensive Cancer Network guidelines do not endorse escalation of chemotherapy for Deauville score 3.
- c) is not appropriate. The RATHL trial demonstrated that bleomycin could be deleted for cycles 3 to 6 of ABVD.
- d) is not appropriate. That would be too much chemotherapy. The EORTC/GELA (Groupe d'Etude des Lymphomes de l'Adulte) H9-U trial⁴ showed that ABVD ×4 RT was noninferior to ABVD ×6 plus RT.
- e) reflects the RATHL trial, demonstrating the equivalence of ABVD ×4 and AVD ×4 after ABVD ×2. However, no RT was included; thus, the value of RT could not be addressed.
- b) is best. After the fourth cycle of ABVD, repeat PET. For Deauville scores 1 to 3, proceed with involved site RT to 30 Gy. This is similar to the protocols from the H9-U trial and Eastern Cooperative Oncology Group E2496 study,⁵ although neither incorporated interim PET to select responding patients for continuation of ABVD. The

inclusion of RT will minimize the risk of relapse (5% with combined modality vs 10% with ABVD alone in H10-U). The RT technique should address both breast and cardiac exposures. The breast V4 should be kept to a minimum and the mean heart dose to <5 Gy. Appropriate intensity modulated RT plans or proton therapy could achieve these goals. Ideally, however, a conversation with the patient (especially if younger than 30) would take place to discuss the relative risks of breast cancer with RT versus higher risk of relapse with ABVD alone.

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Deauville Dubiosity



We would choose option b, 4 cycles of adriamycin, bleomycin, vinblastine, and dacarbazine (ABVD) followed by involved-site radiation therapy to 30 to 36 Gy.

The patient has unfavorable disease with multiple sites of involvement and bulk, which we define as >7 cm.^{1,2} At our institution, we would treat her on a phase 2 protocol of brentuximab vedotin and adriamycin, vinblastine, and dacarbazine, followed by reduced dose and volume radiation therapy (RT).³ Off protocol, we favor the European Organisation for Research and Treatment of Cancer H10 approach, which showed that patients who received combined-modality treatment had improved progression-free survival compared with ABVD $\times$ 4 only.⁴

1. We generally consider Deauville 1 to 3 to be positron emission tomography (PET)-negative; however, PET-negativity was defined heterogeneously on various trials. For example, on the UK RAPID trial, designed to examine the avoidance of RT in patients with early-stage, favorable Hodgkin lymphoma, Deauville 3 after ABVD $\times$ 3 was considered PET-positive, and those patients received RT.
2. All patients need a comprehensive risk-benefit assessment. It is important to understand the patient's risk of relapse and weigh the toxicity of subsequent treatments, including high-dose chemotherapy and autologous stem cell transplant, against the risk of 30 Gy consolidative RT.
3. Depending on the patient's anatomy and the anatomy of the disease, deep-inspiration breath-hold and taping the breasts laterally or using the breast board can significantly limit dose to the heart and breasts.

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Would Negative Interim PET Be Enough to Omit Radiation in Patients With Early Hodgkin Lymphoma? The Answer Is No



The patient in question, with both B symptoms and bulky disease, belongs to an intermediate group in which disease can be considered advanced and treated with bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and prednisone if we follow the German Hodgkin's Study HD 11 criteria.¹ Many others still treat this group as an early unfavorable disease using doxorubicin, bleomycin, vinblastine, and dacarbazine (ABVD). This is particularly important in explaining the treatment options presented here. The interim positron emission tomography (iPET) is practically normal (Deauville scores of 1, 2, or 3 are equivalent to complete response). Based on studies looking at iPET, for European Organisation for Research and Treatment of Cancer 10 unfavorable disease, using adaptive therapy and omitting radiation therapy according to iPET in patients treated with ABVD-like chemotherapy had a failure rate of up to 16%. Thus, both options (e) and (c) are unacceptable.

On the other hand, option (a) would not apply based on the German Hodgkin Study Group 14 study,² which that did not show an overall survival when comparing 4 ABVD to 2 ABVD + 2 bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and prednisone. Additionally, iPET did not guide the treatment decision.

So, now we have eliminated the options with no radiation therapy, leaving us to decide between 4 and 6 ABVD. In view of the negative iPET, based on a recent

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study by Simontacchi et al³ showing a worse outcome for iPET+ disease and suggesting that iPET– disease might do better, and the fact that involved site radiation therapy will follow chemotherapy to consolidate, option (b) is the most reasonable, although option (d) can also be considered.

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Follow Your Heart



This young patient could be classified as having unfavorable early-stage Hodgkin lymphoma (HL) per the European Organization for Research and Treatment of Cancer H10 study or advanced HL per UK Response-Adapted Therapy for Advanced Hodgkin Lymphoma study, both adaptive therapy trials commencing with 2 cycles of Doxorubicin, bleomycin, vinblastine, and dacarbazine.¹

Deauville scores are determined visually and not through standardized uptake value measurements. Therefore, I would review the positron emission tomography/

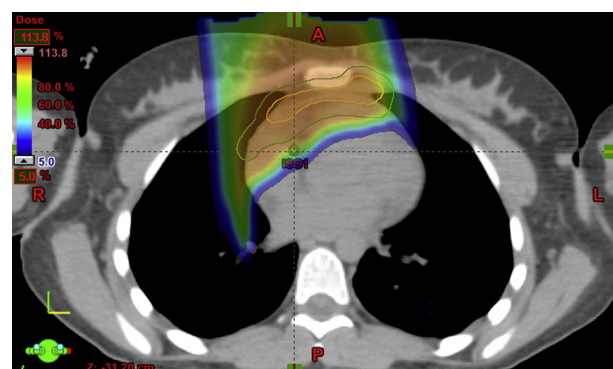


Fig. 1. Colorwash dose distribution for a proton plan.

computed tomography scan with a qualified nuclear radiologist before determining the treatment regimen of choice, whether (1) 2 more cycles of ABVD × 2 + involved-site radiation therapy (ISRT) to 30 to 36 Gy or (2) 4 cycles of Doxorubicin, vinblastine, and dacarbazine. Evaluation of prechemotherapy scans for disease distribution can help determine radiation therapy dose to the heart, lung, and breast tissue with ISRT.

Modern radiation therapy techniques—including deep-inspiration breath-hold, volumetric modulated arc therapy, and proton therapy (Figure 1)—can typically achieve organ-at-risk goals in these cases.² If the expected breast dose is low and mean heart dose is <15 Gy, I would favor completing an additional ABVD × 2 + ISRT to 30 to 36 Gy. I would only consider escalate Bleomycin, Etoposide, Doxorubicin, Cyclophosphamide, Vincristine, Procarbazine, Prednisone after determining a Deauville score of 4 or 5. For a significant breast (>7 Gy, V4 > 20%) or heart dose >15 Gy, 4 cycles of AVD per Response-Adapted Therapy for Advanced Hodgkin Lymphoma study without ISRT is a reasonable option to avoid risks of cardiotoxicity and secondary cancers.^{2,3}

It is of utmost importance to discuss these options with patients so they can decide their priorities in balancing the risk of toxicity with improved disease control.

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